

Simple regression and OLS

Slide route: Lecture3v2.tex: regression line through large-sample distribution. Stock and Watson Chapter 4.

Session 4 | A line, an estimator, an assumption

Outcomes: Compute an OLS line, distinguish errors from residuals, interpret slope units, and state the role of conditional mean zero.

Opening A line can summarize an association. What extra argument would let us call its slope a causal effect? Keep that question on the board throughout the algebra.

100-minute route

0–10	Retrieve covariance and conditional expectation.
10–25	Population model, fitted line, errors versus residuals.
25–45	OLS objective and the three-observation example.
45–55	Interpretation and pause.
55–70	R^2 , SER, fit versus causality.
70–90	Causal assumptions; unbiasedness and sampling variation.
90–100	Exit ticket; hand off inference to Lecture 4.

Slide-by-slide script

Slide 1 | Title

Guide Open with the question: what does the slope of a line summarize, and when could it be causal?

Slide 2 | Roadmap

Guide Preview the sequence: population model, OLS calculation, interpretation, fit, and causal assumptions.

Slide 3 | Last lesson

Guide Retrieve the difference between a population parameter and a sample statistic. Ask where the sampling variation enters.

Slide 4 | Last lesson

Guide Use the conditional-mean table to remind students that conditioning can change the average outcome.

Slide 5 | The regression line

Guide Draw a scatterplot and a line. Ask what feature of the cloud the slope is trying to summarize.

Slide 6 | The regression line

Guide Explain that the fitted line predicts the conditional mean only under the population linear regression interpretation.

Slide 7 | Population linear regression model

Guide Write $Y_i = \beta_0 + \beta_1 X_i + u_i$ and label the systematic part and the error.

Slide 8 | OLS estimator

Guide Introduce the sum of squared residuals. Example: compare candidate intercepts for data 1, 2, 6.

Slide 9 | OLS estimator

Guide Differentiate the objective to obtain the first-order conditions. Circle $\sum \hat{u}_i = 0$ and $\sum X_i \hat{u}_i = 0$.

Slide 10 | Back to the model

Guide Distinguish the unobserved population error u_i from the computed residual $\hat{u}_i$.

Slide 11 | Interpretation

Guide Interpret the slope in units. For test score on class size, a slope of -2.3 means 2.3 fewer points per additional student, as an association.

Slide 12 | Goodness of fit

Guide Define fitted values and residuals. Ask whether a small residual proves a causal relationship.

Slide 13 | R^2

Guide Calculate $R^2 = 1 - SSR/TSS$ using the three-point example. Explain that in-sample fit is not identification.

Slide 14 | SER

Guide Compute $SER = \sqrt{SSR/(n-2)}$ for a simple regression with an intercept. Explain its units.

Slide 15 | Goodness of fit again

Guide Use the same data to show that high R^2 can coexist with a noncausal relation.

Slide 16 | Conditions for causality

Guide List the causal question, a credible source of variation, and the assumptions linking the variation to the effect.

Slide 17 | Conditions for causality

Guide Ask students to name one omitted determinant of test scores correlated with class size. Keep the answer on the board.

Slide 18 | Conditional mean zero

Guide Interpret $\mathbb{E}(u_i | X_i) = 0$ as no systematic omitted determinants left in the error at any X .

Slide 19 | Conditional mean zero

Guide Use a simple violation: wealth affects both class size and test resources. Ask which direction of bias is plausible.

Slide 20 | i.i.d. draws

Guide Explain what independent and identically distributed mean separately. Give a warning about repeated observations from the same school.

Slide 21 | No large outliers

Guide Show why a single extreme X can dominate the denominator in the slope formula. Ask whether dropping it is automatically legitimate.

Slide 22 | OLS sample properties

Guide Separate finite-sample unbiasedness from consistency and large-sample normality.

Slide 23 | Expected slope

Guide Use $\hat{\beta}_1 - \beta_1 = \sum(X_i - \bar{X})u_i / \sum(X_i - \bar{X})^2$ to point to the source of bias.

Slide 24 | Variance of slope

Guide Explain that more spread in X improves precision, while more noise in u reduces it.

Slide 25 | Variance calculation

Guide Keep the algebra secondary. Ask students to identify the numerator and denominator intuitively.

Slide 26 | Variance calculation

Guide Use a units check: changing X from students to tens of students changes the numerical slope and its standard error consistently.

Slide 27 | Large-sample distribution

Guide Connect the CLT idea to a regression coefficient: standardized estimation error becomes approximately normal under assumptions.

Slide 28 | Material

Guide Point students to the chapter and assign the three-point OLS calculation as the first practice problem.

Board example | Compute, do not memorize

Use with slides Lecture3v2 slides 4–9: The regression line, Population Linear Regression Model, and OLS Estimator. Begin the three-point calculation when slide 7 introduces OLS; use slides 8–9 for the first-order conditions and residual identities.

Use synthetic observations $(x, y) = (1, 2), (2, 3), (3, 5)$.

$$\bar{x} = 2, \quad \bar{y} = 10/3, \quad \hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{3}{2} = 1.5.$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 1/3.$$

Predictions are $11/6, 10/3, 29/6$; residuals are $1/6, -1/3, 1/6$. Their sum is zero, and $\sum x_i \hat{u}_i = 0$.

Derivation cue Differentiate $\sum (y_i - b_0 - b_1 x_i)^2$ with respect to both parameters. The first equation puts the fitted line through $(\bar{x}, \bar{y})$; substitute it into the second to obtain the slope formula.

Distinction Population u_i is unobserved. Sample $\hat{u}_i = y_i - \hat{y}_i$ is computed. OLS residual orthogonality is an algebraic identity, not evidence of exogeneity.

Use the same example | Fit and interpretation

Use with slides Lecture3v2 slides 11–15: Interpretation, Goodness of Fit, R^2 , SER, and Goodness of Fit again. Use the same three observations to compute R^2 on slide 13 and SER on slide 14.

$$SSR = \sum \hat{u}_i^2 = 1/6, \quad TSS = \sum (y_i - \bar{y})^2 = 14/3.$$

Here SSR means residual sum of squares; always define the notation because some texts use it for regression sum of squares.

$$R^2 = 1 - \frac{1/6}{14/3} = \frac{27}{28} \simeq .9643, \quad SER = \sqrt{\frac{1/6}{3-2}} = \sqrt{1/6} \simeq .408.$$

These three observations demonstrate arithmetic, not reliable asymptotic inference.

Units exercise If y is a test score and x is class size, slope 1.5 means an associated 1.5-point difference per additional student. If x is divided by 10, the numerical slope multiplies by 10. Predictions do not change.

Causality and sampling

Use with slides Lecture3v2 slides 16–26: conditions for causality, conditional mean zero, i.i.d. draws, outliers, and OLS sample properties. Use the bias expression beside slides 18–19 before moving to large-sample distribution.

For $Y_i = \beta_0 + \beta_1 X_i + u_i$, a zero conditional mean $\mathbb{E}(u_i | X_i) = 0$ says omitted determinants average to zero at each regressor value. Richer districts may have both different class sizes and different resources, violating this condition.

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (X_i - \bar{X}) u_i}{\sum_i (X_i - \bar{X})^2}.$$

Use the expression to explain why variation in X is needed and why correlation between X and omitted determinants is a problem. Discuss random sampling and finite-moment conditions before invoking large-sample normality.

Side question Does a high R^2 establish causality? No. Does a low R^2 rule out a causal effect? No. Fit and identification answer different questions.

Exit answers Residuals sum to zero when an intercept is included. That fact cannot verify $\mathbb{E}(u | X) = 0$. The intercept may have no sensible substantive interpretation when $X = 0$ is outside the observed range.

If short / preparation Assign the detailed asymptotic variance algebra for review. Preserve the numerical OLS calculation and exogeneity discussion. Read Chapter 5 before session 5.

Source: Lecture3v2.tex; synthetic three-observation calculation.